

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

Answers will vary, but all traffic to the 192.168.2.0/24 network from R3 will be routed through R1. The S0/0/1 interface on R2 is still configured as a passive interface, so OSPF routing information is not being advertised across this interface. The 129 accumulated cost results from the fact that traffic from R3 to the 192.168.2.0/24 network must pass through two T1 (1.544 Mb/s) serial links (at a cost of 64 each) plus the R2 Gigabit 0/0 LAN link (at a cost of 1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2 (config # router ospf 1)

R2 (config-router) # no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial10/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. Add the default serial link cost (64) from R3 to R2 to the Gigabit Ethernet LAN cost (1) on R2.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

% OSPF: Reference bandwidth is changed.

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

In order to enable OSPF to distinguish costs between high-speed connections (such as GigabitEthernet and 10GbE), as the default value of 100 Mbps reference limits all faster connections to cost 1.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

To 192.168.23.0/30: R1 S0/0/1 + R3 G0/0 ($781+1=782$)

To 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 ($781+64=845$)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network travels over two serial links. $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0(781) + R2-S0/0/1(781) + R3-G0/0(1) = 1563. This metric is smaller than the accumulated cost of R1-S0/0/1(1565) + R3-G0/0(1) = 1566

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router ID assignments control the Designated Router (DR) and Backup Designated Router (BDR) election process on a multi-access network. If the router ID is associated with an active interface, it can change if the interface goes down. For this reason, it should be set using the IP address of a loopback interface (which cannot go down) or set using the router-id command.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election process is only an issue on a multi-access network such as Ethernet or Frame Relay. The serial links used in this lab are point-to-point links, so no DR/BDR election is performed.

3. Why would you want to set an OSPF interface to passive?

Answers will vary, but configuring a LAN interface as passive eliminated unnecessary OSPF routing information on that interface, freeing up bandwidth. The router will still advertise the network to its neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of OSPF Areas. Dividing the network into smaller areas localizes routing updates, keeps routing tables small, and reduces the router's CPU and memory usage.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route. Routers choose the path with the lowest Administrative Distance (AD), ignoring the metrics. EIGRP (AD 90) is preferred over OSPF (AD 110) and RIP (AD 120.)